Experiment 18: AgriMart – Online Agricultural Shopping System

Aim:

To develop an interactive agricultural e-commerce website where farmers can manage products and orders, while customers can search agricultural supplies, manage their wallet and cart, place orders, and track shipping status.

Algorithm:

1. Design a responsive homepage displaying agricultural products such as seeds, fertilizers, pesticides, and farming tools.
2. Implement farmer and customer registration, login, and simulated OTP verification.
3. Allow farmers to add, edit, delete, and manage products, view sales, and update order and shipping statuses.
4. Allow customers to search and filter products, view product details, and add, edit, or remove cart items.
5. Provide wallet recharge, shipping-address management, and order placement with wallet-balance validation.
6. Generate order IDs, save account and order information in browser storage, and display updated order and shipping statuses.

Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>AgriMart | Farm Supplies</title>

<style>

:root{--green:#176b43;--lime:#d9f99d;--bg:#f4f7f1;--ink:#19372a}

*{box-sizing:border-box}
```

```
body{margin:0;background:var(--bg);font-family:Arial,sans-serif;color:var(--ink)}
header{background:var(--green);color:white;padding:17px 5%;display:flex;align-items:center;justify-content:space-between;gap:12px;flex-wrap:wrap}
.logo{font-size:25px;font-weight:800}
.logo span{color:var(--lime)}
button,select,input,textarea{font:inherit;padding:10px;border:1px solid #d6dfd5;border-radius:9px}
button{cursor:pointer;background:var(--green);color:white;border:0;font-weight:bold}
button:hover{filter:brightness(1.12)}
button.secondary{background:#e5eee4;color:var(--green)}
button.danger{background:#b83b36}
nav{display:flex;gap:7px;flex-wrap:wrap}
nav button{background:#ffffff20}
nav button.active{background:var(--lime);color:var(--green)}
main{max-width:1200px;margin:24px auto;padding:0 18px}
.hero{background:linear-gradient(115deg,#174d32,#378451);color:white;border-radius:20px;padding:30px;display:flex;justify-content:space-between;gap:20px;align-items:center;flex-wrap:wrap}
.hero h1{font-size:clamp(27px,4vw,44px);margin:0 0 12px}
.hero p{max-width:580px;line-height:1.6}
.hero .emoji{font-size:75px}
.stats{display:grid;grid-template-columns:repeat(auto-fit,minmax(155px,1fr));gap:12px;margin:20px 0}
.stat,.panel,.product{background:white;border:1px solid #e4eae1;border-radius:15px;padding:17px}
.stat small{color:#718075}.stat strong{display:block;font-size:25px;margin-top:8px}
.toolbar{display:flex;gap:10px;flex-wrap:wrap;margin:20px 0}
.toolbar input{flex:1;min-width:180px}
.products{display:grid;grid-template-columns:repeat(auto-fit,minmax(220px,1fr));gap:15px}
```

```
.product .pic{background:#eef5e8;border-radius:12px;text-align:center;font-size:55px;padding:18px}

.product h3{margin-bottom:5px}

.product p{color:#647369;line-height:1.5;font-size:14px}

.price{font-size:22px;font-weight:bold;color:var(--green)}

.product button{width:100%;margin-top:10px}

.tag{display:inline-block;background:#e6f3dc;color:#28623c;border-radius:20px;padding:5px 9px;font-size:12px}

.panel{margin:15px 0}

.panel h2{margin-top:0}

.formgrid{display:grid;grid-template-columns:repeat(auto-fit,minmax(190px,1fr));gap:10px}

.formgrid input,.formgrid select,.formgrid textarea{width:100%;min-width:0}

table{width:100%;border-collapse:collapse;min-width:570px}

th,td{text-align:left;padding:12px;border-bottom:1px solid #e5ebe2}

th{background:#eef5e8}

.tablewrap{overflow-x:auto}

.row{display:flex;gap:8px;align-items:center;flex-wrap:wrap}

.muted{color:#718075}

.notice{padding:12px;background:#eaf6df;border-radius:10px;margin:12px 0}

footer{text-align:center;color:#718075;padding:25px}

.hidden{display:none!important}

@media(max-width:550px){header{align-items:flex-start}.hero{padding:22px}.hero .emoji{font-size:45px}main{padding:0 12px}}

</style>

</head>

<body>

<header>

 <div class="logo">🌱 Agri<span>Mart</span></div>
```

```
 <nav>
  <button class="active" onclick="setRole('customer',this)">Customer</button>
  <button onclick="setRole('farmer',this)">Farmer</button>
  <button onclick="setRole('account',this)">My Account</button>
  <button onclick="setRole('orders',this)">My Orders</button>
 </nav>
</header>

<main>
<section class="hero">
 <div>
  <span class="tag">YOUR DIGITAL FARM STORE</span>
  <h1>Grow Better.<br>Harvest More.</h1>
  <p>Shop quality seeds, fertilizers, crop protection products and modern farming tools—all in one place.</p>
  <button style="background:#d9f99d;color:#174d32" onclick="document.getElementById('catalog').scrollIntoView({behavior:'smooth'})">Explore Products →</button>
 </div>
 <div class="emoji">🌾 🚜</div>
</section>

<div class="stats">
 <div class="stat"><small>Available Products</small><strong id="productCount">0</strong></div>
 <div class="stat"><small>Cart Items</small><strong id="cartCount">0</strong></div>
 <div class="stat"><small>My Orders</small><strong id="orderCount">0</strong></div>
 <div class="stat"><small>Wallet Balance</small><strong id="walletStat">₹0</strong></div>
```

```
</div>


<section id="catalog">

<h2> 🌿 Agricultural Products</h2>

<div class="toolbar">

 <input id="search" placeholder="Search seeds, tools, fertilizer..."
oninput="renderProducts()">

 <select id="category" onchange="renderProducts()">

  <option value="All">All Categories</option>


<option>Seeds</option><option>Fertilizers</option><option>Pesticides</option><opti
on>Equipment</option>

 </select>

 <select id="sort" onchange="renderProducts()">

  <option value="featured">Featured</option>

  <option value="low">Price: Low to High</option>

  <option value="high">Price: High to Low</option>

 </select>

</div>

<div id="products" class="products"></div>

</section>


<section id="customerPanel" class="panel">

<h2> 🛒 Shopping Cart & Checkout</h2>

<div id="cart"></div>

<div class="formgrid" style="margin-top:15px">

 <input id="customerName" placeholder="Your name">

 <input id="customerPhone" placeholder="Phone number">

 <input id="address" placeholder="Full shipping address">
```

```
</div>
<div class="row" style="margin-top:12px">
 <strong id="total">Total: ₹0</strong>
 <button onclick="checkout()">Place Order with Wallet</button>
</div>
<p class="muted">Demo checkout uses your local browser wallet; no real payment is processed.</p>
</section>

<section id="accountPanel" class="panel hidden">
<h2>👤 Customer Account</h2>
<p>Demo customer account and wallet management.</p>
<div class="formgrid">
 <input id="accountName" placeholder="Customer name">
 <input id="accountPhone" placeholder="Phone number">
</div>
<div class="row" style="margin-top:12px">
 <button onclick="saveAccount()">Save Account</button>
 <input id="recharge" type="number" min="1" placeholder="Recharge amount ₹">
 <button onclick="rechargeWallet()">Add Wallet Money</button>
</div>
<div class="notice" id="accountInfo"></div>
</section>

<section id="ordersPanel" class="panel hidden">
<h2>📦 My Orders & Shipping Tracker</h2>
<div id="orders"></div>
</section>
```

```
<section id="farmerPanel" class="panel hidden">

<h2> 🚜 Farmer Dashboard</h2>

<div class="notice">Demo farmer workspace. Add products, maintain listings, and update order and shipping status.</div>

<div class="stats">

 <div class="stat"><small>My Listings</small><strong id="farmerProducts">0</strong></div>

 <div class="stat"><small>Orders to Manage</small><strong id="farmerOrders">0</strong></div>

 <div class="stat"><small>Total Sales</small><strong id="sales">₹0</strong></div>

</div>


<h3>Add a Product</h3>

<div class="formgrid">

 <input id="pname" placeholder="Product name">

 <select id="pcategory"><option>Seeds</option><option>Fertilizers</option><option>Pesticides</option><option>Equipment</option></select>

 <input id="pprice" type="number" min="1" placeholder="Price ₹">

 <input id="pstock" type="number" min="0" placeholder="Stock quantity">

 <input id="pimage" placeholder="Product image URL (optional)">

 <input id="pdescription" placeholder="Short description">

</div>

<div class="row" style="margin-top:12px">

 <button onclick="addProduct()">Add Product</button>

 <button class="secondary" onclick="resetProducts()">Reset Demo Products</button>

</div>
```

<h3>Manage My Products</h3>

<h3>Manage Orders & Shipping</h3>

</section>

</main>

<footer>AgriMart · Internet Programming Lab Experiment 18 · Educational Demo</footer>

<script>

const initialProducts=[

{id:1,name:"Premium Rice Seeds",category:"Seeds",price:450,stock:30,emoji:" 🌾 ",description:"Quality seeds for healthy crop growth."},

{id:2,name:"Organic Compost",category:"Fertilizers",price:320,stock:25,emoji:" 🌱 ",description:"Organic nutrients to enrich soil."},

{id:3,name:"Drip Irrigation Kit",category:"Equipment",price:1250,stock:12,emoji:" 💧 ",description:"Water-saving irrigation equipment."},

{id:4,name:"Neem Crop Spray",category:"Pesticides",price:280,stock:18,emoji:" 🌿 ",description:"Neem-based crop protection product."},

{id:5,name:"Vegetable Seed Pack",category:"Seeds",price:180,stock:40,emoji:" 🥬 ",description:"A selection of vegetable seeds."},

{id:6,name:"Hand Garden Tools",category:"Equipment",price:550,stock:15,emoji:" 🧰 ",description:"Useful tools for everyday farm work."}

];

function load(key,fallback){try{return JSON.parse(localStorage.getItem(key))??fallback}catch{return fallback}}

```
let products=load("agri_products",initialProducts);
let cart=load("agri_cart",[]);
let orders=load("agri_orders",[]);
let wallet=Number(localStorage.getItem("agri_wallet")||1500);
let account=load("agri_account",{name:"",phone:""});
let role="customer",editingId=null;

function save(){
 localStorage.setItem("agri_products",JSON.stringify(products));
 localStorage.setItem("agri_cart",JSON.stringify(cart));
 localStorage.setItem("agri_orders",JSON.stringify(orders));
 localStorage.setItem("agri_wallet",String(wallet));
 localStorage.setItem("agri_account",JSON.stringify(account));
}
function money(n){return "₹"+Number(n).toLocaleString("en-IN")}
function escapeHTML(s){return
String(s).replace(/[&<>"']/g,c=>({"&":"&amp;","<":"&lt;",">":"&gt;","'":"&quot;","'":"&#39;"}[c
]))}
function setRole(next,btn){
 role=next;
 document.querySelectorAll("nav button").forEach(b=>b.classList.remove("active"));
 if(btn)btn.classList.add("active");

["customerPanel","accountPanel","ordersPanel","farmerPanel"].forEach(id=>document.
getElementById(id).classList.add("hidden"));

if(next==="customer")document.getElementById("customerPanel").classList.remove("
hidden");
```

```
if(next==="account")document.getElementById("accountPanel").classList.remove("hid
den");

if(next==="orders")document.getElementById("ordersPanel").classList.remove("hidden
");

if(next==="farmer")document.getElementById("farmerPanel").classList.remove("hidden
");
 renderAll();
}
function renderProducts(){
 let q=document.getElementById("search").value.toLowerCase();
 let cat=document.getElementById("category").value;
 let list=products.filter(p=>(p.name+" "+p.description+"
"+p.category).toLowerCase().includes(q)&&(cat==="All"||p.category===cat));
 let sort=document.getElementById("sort").value;
 if(sort==="low")list.sort((a,b)=>a.price-b.price);
 if(sort==="high")list.sort((a,b)=>b.price-a.price);
 document.getElementById("products").innerHTML=list.length?list.map(p=>`
 <article class="product">
 <div class="pic">${p.image?`<img src="${escapeHTML(p.image)}"
alt="${escapeHTML(p.name)}" style="width:100%;height:100px;object-
fit:contain">`:p.emoji||"🌿"}</div>
 <p><span class="tag">${escapeHTML(p.category)}</span></p>
 <h3>${escapeHTML(p.name)}</h3><p>${escapeHTML(p.description)}</p>
 <p class="muted">Stock: ${p.stock}</p><div class="price">${money(p.price)}</div>
 <button ${p.stock<1?"disabled":""} onclick="addToCart(${p.id})">${p.stock<1?"Out of
Stock":"Add to Cart +"}</button>
 </article>`).join(""):'<div class="panel">No products found. Try another search.</div>';
 document.getElementById("productCount").textContent=products.length;
```

```
}
function addToCart(id){
 let p=products.find(x=>x.id===id);if(!p||p.stock<1)return alert("Product is out of stock.");
 let item=cart.find(x=>x.id===id);
 if(item){if(item.qty>=p.stock)return alert("Maximum available stock reached.");item.qty++}
 else cart.push({id,qty:1});
 save();renderAll();
}
function changeQty(id,delta){
 let item=cart.find(x=>x.id===id),p=products.find(x=>x.id===id);
 if(!item)return;
 if(delta>0&&item.qty>=p.stock)return alert("Maximum available stock reached.");
 item.qty+=delta;if(item.qty<=0)cart=cart.filter(x=>x.id!==id);
 save();renderAll();
}
function removeCart(id){cart=cart.filter(x=>x.id!==id);save();renderAll()}
function renderCart(){
 let total=0;
 document.getElementById("cart").innerHTML=cart.length?cart.map(item=>{
 let p=products.find(x=>x.id===item.id);if(!p)return "";
 total+=p.price*item.qty;
 return `<div class="row" style="justify-content:space-between;border-bottom:1px solid #eee;padding:10px 0">
 <span><b>${escapeHTML(p.name)}</b><br>${money(p.price)} each</span>
 <span class="row"><button class="secondary" onclick="changeQty(${p.id},-1)">−</button>${item.qty}<button class="secondary" onclick="changeQty(${p.id},1)">+</button><button class="danger" onclick="removeCart(${p.id})">Remove</button></span>
```

```
</div>`}).join(""):'<p class="muted">Your cart is empty. Add products to begin.</p>';
 document.getElementById("total").textContent="Total: "+money(total);
}
function checkout(){
 if(!cart.length)return alert("Your cart is empty.");
 let name=document.getElementById("customerName").value.trim()||account.name;
 let phone=document.getElementById("customerPhone").value.trim()||account.phone;
 let address=document.getElementById("address").value.trim();
 if(!name||!phone||!address)return alert("Enter your name, phone number, and shipping address.");
 for(let item of cart){let p=products.find(x=>x.id===item.id);if(!p||p.stock<item.qty)return alert("Insufficient stock for "+(p?.name||"a product"))}
 let total=cart.reduce((s,item)=>s+products.find(p=>p.id===item.id).price*item.qty,0);
 if(total>wallet)return alert("Insufficient wallet balance. Please recharge your wallet.");
 wallet-=total;
 let id="AG"+Date.now().toString().slice(-8);

orders.unshift({id,name,phone,address,items:cart.map(i=>({...i})),total,status:"Confirmed",shipping:"Processing",date:new Date().toLocaleString()});
 cart.forEach(item=>{let p=products.find(x=>x.id===item.id);p.stock-=item.qty});
 account={...account,name,phone};cart=[];save();renderAll();
 alert("Order placed successfully! Order ID: "+id);
}
function saveAccount(){
 account.name=document.getElementById("accountName").value.trim();
 account.phone=document.getElementById("accountPhone").value.trim();
 if(!account.name||!account.phone)return alert("Enter your name and phone number.");
 save();renderAll();alert("Demo account saved.");
}
```

```
function rechargeWallet(){
 let amount=Number(document.getElementById("recharge").value);
 if(!Number.isFinite(amount)||amount<=0)return alert("Enter a valid recharge amount.");

wallet+=amount;document.getElementById("recharge").value="";save();renderAll();alert("Demo wallet recharged by "+money(amount));
}
function renderOrders(){
 document.getElementById("orders").innerHTML=orders.length?orders.map(o=>`
 <div class="panel">
 <div class="row" style="justify-content:space-between"><h3>Order ${escapeHTML(o.id)}</h3><span class="tag">${escapeHTML(o.status)}</span></div>
 <p>${escapeHTML(o.date)} · Total <b>${money(o.total)}</b></p>
 <p>Shipping: <b>${escapeHTML(o.shipping)}</b></p>
 <p>Address: ${escapeHTML(o.address)}</p>
 <p>${o.items.map(i=>{let p=products.find(x=>x.id===i.id);return escapeHTML(p?.name||"Removed product")+" × "+i.qty}).join(", ")}</p>
 </div>`).join(""):'<p class="muted">No orders yet.</p>';
}
function addProduct(){
 let name=document.getElementById("pname").value.trim(),price=Number(document.getElementById("pprice").value),stock=Number(document.getElementById("pstock").value);
 if(!name||!Number.isFinite(price)||price<=0||!Number.isInteger(stock)||stock<0)return alert("Enter a product name, valid price, and non-negative whole-number stock.");
 let category=document.getElementById("pcategory").value;
 let image=document.getElementById("pimage").value.trim();
 let description=document.getElementById("pdescription").value.trim()||"Fresh agricultural supply.";
```

```
products.unshift({id:Date.now(),name,category,price,stock,emoji:({Seeds:" 🌾 ",Fertilizers:" 🌱 ",Pesticides:" 🍃 ",Equipment:" 🧰 "})[category],image,description});

["pname","pprice","pstock","pimage","pdescription"].forEach(id=>document.getElementById(id).value="");
 save();renderAll();alert("Product added.");
}
function deleteProduct(id){
 if(!confirm("Delete this product listing?"))return;
 products=products.filter(p=>p.id!==id);cart=cart.filter(i=>i.id!==id);save();renderAll();
}
function editProduct(id){
 let p=products.find(x=>x.id===id);if(!p)return;
 let name=prompt("Product name:",p.name);if(name===null)return;
 let price=Number(prompt("Price:",p.price));if(!Number.isFinite(price)||price<=0)return alert("Invalid price.");
 let stock=Number(prompt("Stock quantity:",p.stock));if(!Number.isInteger(stock)||stock<0)return alert("Invalid stock.");
 p.name=name.trim()||p.name;p.price=price;p.stock=stock;save();renderAll();
}
function updateStatus(id,field,value){
 let o=orders.find(x=>x.id===id);if(!o)return;
 o[field]=value;save();renderAll();
}
function resetProducts(){
 if(!confirm("Reset products to the original demo catalogue?"))return;
 products=initialProducts.map(p=>({...p}));save();renderAll();
}
function renderFarmer(){
```

```
 document.getElementById("farmerProducts").textContent=products.length;

 document.getElementById("farmerOrders").textContent=orders.length;


document.getElementById("sales").textContent=money(orders.filter(o=>o.status!=="Cancelled").reduce((s,o)=>s+o.total,0));

 document.getElementById("farmerProductList").innerHTML=products.length?`<div class="tablewrap"><table><tr><th>Product</th><th>Price</th><th>Stock</th><th>Actions</th></tr>${products.map(p=>`<tr><td>${escapeHTML(p.name)}</td><td>${money(p.price)}</td><td>${p.stock}</td><td><button onclick="editProduct(${p.id})">Edit</button> <button class="danger" onclick="deleteProduct(${p.id})">Delete</button></td></tr>`).join("")}</table></div>`:"No product listings.";


document.getElementById("farmerOrderList").innerHTML=orders.length?orders.map(o=>`<div class="panel">

 <b>${escapeHTML(o.id)}</b> · ${escapeHTML(o.name)} · ${money(o.total)}

 <p class="muted">${escapeHTML(o.address)}</p>

 <div class="formgrid">

 <label>Order status<select onchange="updateStatus('${o.id}','status',this.value)">${["Confirmed","Processing","Packed","Cancelled"].map(s=>`<option ${o.status===s?"selected":""}>${s}</option>`).join("")}</select></label>

 <label>Shipping status<select onchange="updateStatus('${o.id}','shipping',this.value)">${["Processing","Packed","Shipped","Out for Delivery","Delivered"].map(s=>`<option ${o.shipping===s?"selected":""}>${s}</option>`).join("")}</select></label>

 </div></div>`).join(""):"No orders to manage.";

}

function renderAll(){

 renderProducts();renderCart();renderOrders();renderFarmer();

 document.getElementById("cartCount").textContent=cart.reduce((s,i)=>s+i.qty,0);

 document.getElementById("orderCount").textContent=orders.length;

 document.getElementById("walletStat").textContent=money(wallet);
```

```
 document.getElementById("accountInfo").textContent=`Name: ${account.name||"Not set"} | Phone: ${account.phone||"Not set"} | Wallet: ${money(wallet)}`;

 document.getElementById("accountName").value=account.name;

 document.getElementById("accountPhone").value=account.phone;

}

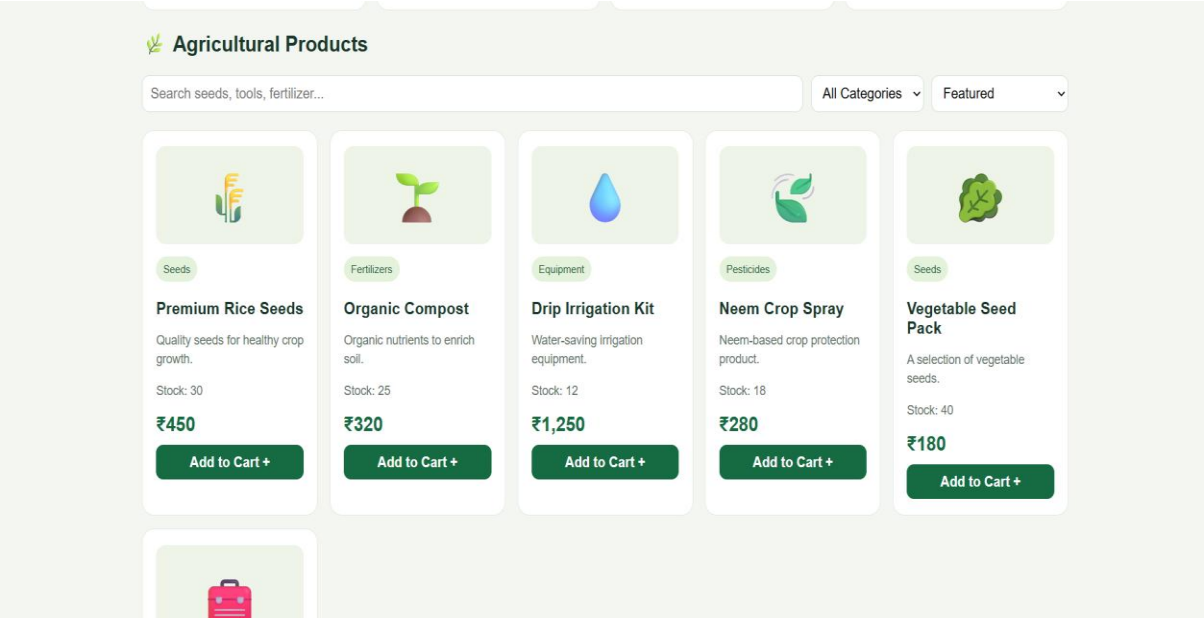
renderAll();

</script>

</body>

</html>
```

Output:



Result:

Thus, the AgriMart online agricultural shopping system was developed successfully with product management, customer cart and wallet management, order placement, and shipping-status tracking.